

Problem 18.10

Adam buys a perpetuity due of \$1,000 per month for \$100,000. Calculate the annual effective rate of interest used to calculate the price of this perpetuity.

Solution

Let j be the effective monthly interest rate. Since the perpetuity is paid at the beginning of each month, it is a perpetuity-due. The present value is

$$PV = \frac{R}{d},$$

where d is the effective monthly discount rate.

Here,

$$PV = 100000 \quad \text{and} \quad R = 1000.$$

Thus,

$$100000 = 1000 \left(\frac{1}{d} \right).$$

So

$$d = \frac{1000}{100000} = 0.01.$$

The relationship between j and d is

$$d = \frac{j}{1+j}.$$

Therefore,

$$0.01 = \frac{j}{1+j}.$$

Solving,

$$0.01(1+j) = j,$$

so

$$0.01 = 0.99j.$$

Hence,

$$j = \frac{1}{99}.$$

The annual effective interest rate is

$$i = (1+j)^{12} - 1.$$

Thus,

$$\begin{aligned} i &= \left(1 + \frac{1}{99} \right)^{12} - 1 \\ &\approx 0.128178. \end{aligned}$$

Therefore, the annual effective rate of interest is

$$\boxed{12.82\%}.$$